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Palaeomagnetic and rock magnetic investigations of the Sipoo Subjotnian quartz porphyry and diabase dykes, southern Fennoscandia

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Abstract

The Sipoo quartz porphyry dyke swarm (U–Pb zircon age 1.63 Ga) in southern Finland carries an intermediate to high coercivity remanent magnetization A_N ($D = 22.2°$, $I = -1.8°$, $k = 34.2$, $\alpha 95 = 13.3°$, $n = 5$ dykes) which is regarded as primary. The Sipoo diabase dykes, emplaced slightly before the granite porphyries, yield a highly scattered antiparallel direction A_R ($D = 198.0°$, $I = -6.5°$, $k = 6.2$, $\alpha 95 = 40.1°$, $n = 4$ dykes) isolated in intermediate to high coercivities. Based on positive contact test results, remanence A_R is primary in the diabases. The best preserved diabase dyke, as determined by petrographic and bulk magnetization properties, carries both the A_R magnetization and the A_N magnetization as a secondary component in some samples. The most altered diabase dyke is almost completely overprinted by A_N. The overprinting has taken place as a result of hydrothermal alteration and formation of new magnetite during the emplacement of quartz porphyry dykes and rapakivi granite. The quartz porphyries and diabases carry mixed SD and MD magnetic grain sizes, the large MD grains being responsible for a relatively large Present Earth's Field (PEF) magnetization. In addition, a high coercivity magnetization B ($D = 23.9°$, $I = 52.2°$, $k = 215.8$, $\alpha 95 = 6.3°$, $n = 4$ dykes), presumably a chemical overprint, has been isolated in the quartz porphyries. The B magnetization is interpreted to be either an intermediate direction between A_N and PEF with no geological significance, or to represent a separate reactivation of the crust at about 1.32 Ga ago.

1. Introduction

Subjotnian time (1600–1300 Ma) in Fennoscandia is characterized by anorogenic magmatism comprising large volumes of rapakivi granites and associated dyke rocks. Palaeomagnetically the Subjotnian is defined by a self-closing loop of the Apparent Polar Wander (APW) Path where the Subjotnian poles are mostly determined from rapakivi granites and mafic dyke swarms (see Pesonen et al., 1991; Elming et al., 1993). In this paper we present palaeomagnetic results from quartz porphyry and diabase dykes associated with the Onas rapakivi granite body in southern Finland. This rapakivi is situated between the rapakivi granites of Wiborg and Åland. The main aim of our palaeomagnetic study was to isolate a Subjotnian remanence from the diabases and quartz porphyry dykes to add more data to the APW Path and thus to improve our knowl-

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edge of the kinematics of the Fennoscandian shield. By petrographic, petrophysical and rock magnetic studies we have tried to resolve the mechanism by which the remanences have been acquired in order to seek their connections to the known geological processes. In addition, we have tried to find out the role of the secondary remanence components of the dykes which can provide evidence of later geological events in southern Fennoscandia.

2. Geological setting

2.1. Regional geology

Southern Finland (Fig. 1) is mostly covered by crust formed during the Svecofennian orogeny about 1.9–1.87 Ma ago (Huhma, 1986). The youngest rock types in the area are the postorogenic rapakivi granites showing U–Pb zircon ages of 1650–1630 Ma in the southeastern part of Finland and youngest ages of 1590–1540 Ma in the southwestern part of Finland (Vaasjoki, 1977; Patchett et al., 1981; Vaasjoki et al., 1991). The Finnish rapakivi granites most probably derive from crust which formed during the Svecofennian orogeny 200–300 Ma earlier (Rämö, 1991; Vaasjoki et al., 1991). Genetically related to the rapakivi granites are the Subjotnian diabase dykes and quartz porphyry dykes which form northwest–southeast-directed dyke swarms in southeastern Finland. The Svecofennian crust was already deeply eroded during the emplacement of the Subjotnian dykes (Rämö, 1990). The most extensive dyke swarm is the about 70 km wide Häme diabase swarm extending about 250 km

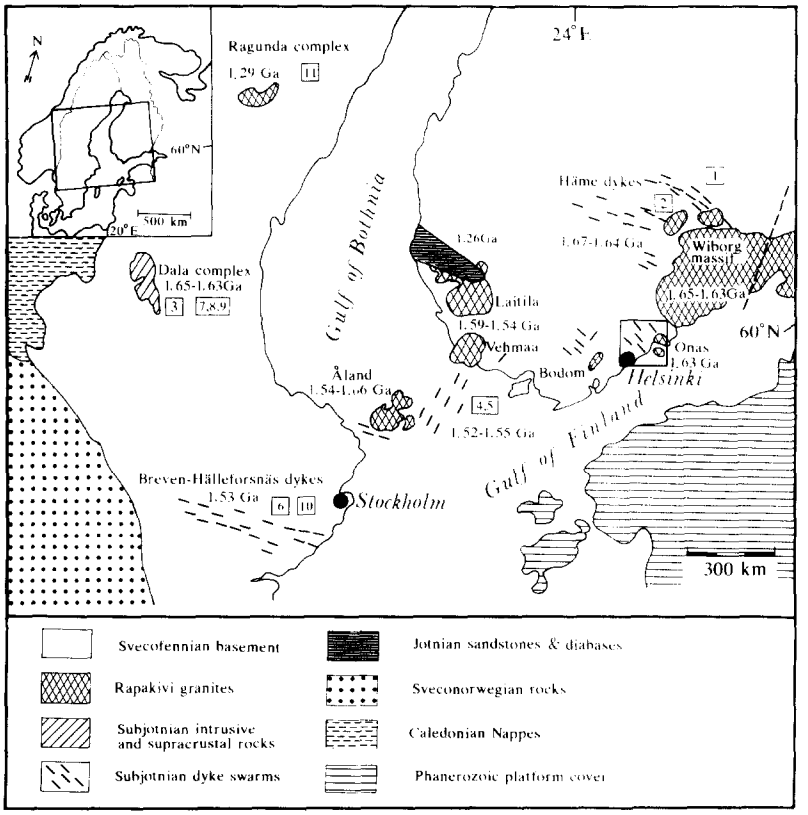


Fig. 1. Simplified geological map of the southern Fennoscandian Shield, according to Rämö (1991) and other sources. The study area is enclosed by a rectangle. Numbers refer to poles in Table 5 and Fig. 17.

from the Wiborg rapakivi granite batholith and smaller associated rapakivi complexes in south-eastern Finland (Fig. 1). Further to the west there are smaller rapakivi plutons, such as the Onas, Bodom and Obbnäs rapakivi granites which all have associated Subjotnian diabase and quartz porphyry dykes. Further west there occur the rapakivi granites of Åland, Laitila and Vehmaa of age 1590–1540 Ma. Rapakivi granites also occur in Estonia and Latvia (Koistinen, 1994) some 50–250 km to the south from Finland. In Sweden the rapakivi granites like Nordingrå and Ragunda appear to be somewhat younger than in Finland. Nordingrå gabbro–anorthosite has U–Pb (Zr) and Rb–Sr ages of 1385–1550 Ma (Welin and Lundqvist, 1970; Kornfält, 1976) and Ragunda rapakivi granite has a Rb–Sr age of 1291 ± 30 Ma (Kornfält, 1976) which is the minimum age of the intrusion. Clearly distinct from Subjotnian magmatic episode is the post-Jotnian one, associated with early Jotnian sandstones (K–Ar age 1130–1318 Ma, Suominen, 1991), cut by post-Jotnian dykes and sills. They have U–Pb (Zr) ages of approximately 1270–1250 Ma (Suominen, 1991).

In the Wiborg area, the whole post-Svecofennian process started by the emplacement of the diabase dykes before about 1665 Ma (Vaasjoki et al., 1991). The diabase emplacement took place in three different pulses at 1665, 1645 and 1635 Ma and possibly peaked at about 1645 Ma (Laitakari and Leino, 1989; Vaasjoki and Sakko, 1989; Vaasjoki et al., 1991). Hence, the emplacement of the diabase dykes took about 30 Ma. In Åland rapakivi area the diabase dykes have U–Pb zircon ages of about 1575 ± 10 Ma (Suominen, 1991). Quartz-feldspar porphyry dykes in the Wiborg rapakivi area register U–Pb zircon ages of 1635 Ma (Vaasjoki et al., 1991) Ma and are thus slightly younger than the rapakivi granite. The last quartz porphyries intruded 1615 ± 5 Ma ago in the Wiborg area (Vaasjoki et al., 1991).

3. Geology and isotopic ages of the Sipoo area

The geology of the Subjotnian Sipoo quartz porphyry dykes has been thoroughly investigated and described by Borgström (1907, 1947), Härme (1978) and Törnroos (1984). The Sipoo diabase and quartz porphyry dykes (Fig. 2) follow the general northwest and west–northwest trend of the dyke sets of southern Finland and cut sharply the Svecofennian bedrock. The width of the dyke swarm is about 80 km and it can be followed over 20 km distance. Dykes are mostly vertical or subvertical. The strike of most quartz porphyry dykes is northwest–southeast, while the direction of the diabases trends more west–east.

The width of the quartz porphyry dykes ranges from some centimetres to 25 m, the typical width is about 3–5 m. The dykes are fine grained with aphanitic contacts (Törnroos, 1984). The main constituents are K-feldspar, quartz and plagioclase and the accessories are biotite, hornblende, apatite, fluorite and magnetite–ilmenite. Primary biotite and amphibole indicate that the magma was hydrous (Törnroos, 1984). The K-feldspar phenocrysts can be up to 2 cm long, while the quartz and plagioclase phenocrysts are smaller. In some quartz porphyry dykes, there are plagioclase and magnetite–ilmenite phenocrysts in an orb-like spherulitic groundmass (Törnroos, 1984).

The width of the diabase dykes ranges from some centimetres to 3 m. The mode of occurrence of separate dykes differ. Dyke SC (Fig. 2) is composed of several, about 1 m wide, parallel branches, while dyke SF is narrow (about 30 cm wide) and it is fractured, especially at the margins. The width of dyke SD could not be determined because it is poorly exposed. The length of the plagioclase phenocrysts in dyke SD is up to 5 cm. In some dykes there is some unaltered augite, but the main minerals of the diabases are hornblende, biotite and plagioclase. Some quartz also exists. Magnetite is a main constituent in some dykes and forms lath-like grains and small aggregates of grains. The texture of the diabases is subophitic.

The U–Pb zircon age of the Sipoo quartz porphyry dykes is 1633 Ma (Törnroos, 1984) corresponding to the age of 1633 ± 10 Ma of the granite porphyry dyke in the Pellinge area further to the southeast (Vaasjoki et al., 1992). In one site around the Onas rapakivi granite, the quartz porphyries form composite dykes together with the almost coeval diabases where the fissure was

first intruded by diabase magma and slightly later the central part of the diabase crack was intruded by quartz porphyry magma (Laitala, 1984). The age of the Onas rapakivi is 1630 ± 10 Ma (Laitala, 1984). Törnroos (1984) has suggested that the quartz porphyry dykes were emplaced slightly before the Onas rapakivi granite in the extensional fractures which were formed as a result of shearing during the Svecofennian orogeny about 1.9–1.8 Ga ago and reopened during the rapakivi intrusion. Laitakari and Leino (1989) have proposed that the dykes were emplaced in rifts which are related to the diapiric ascent of the rapakivi magma. Diabase dykes in the Sipoo area have not been dated isotopically, but because they are almost coeval with the quartz porphyry dykes (see above) their age can be regarded as 1633 Ma.

Chilled margins of the quartz porphyry dykes indicate that the Svecofennian host rock was colder than the dyke magma at the time of intrusion (Törnroos, 1984). According to Törnroos (1984), after the emplacement of the quartz porphyries, the only process that could have caused alterations in the diabase and quartz porphyry

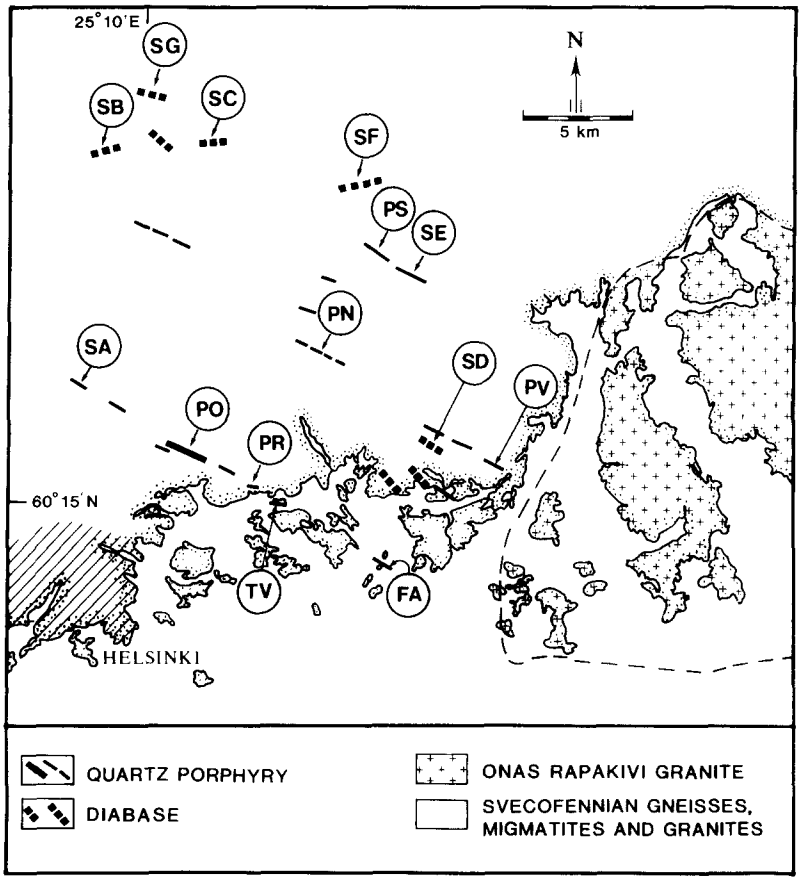


Fig. 2. Geology and sampling sites of the Sipoo area. Modified from Törnroos (1984).

dykes was the heat flow and pore fluids related to the final stages of the crystallization of the Onas rapakivi magma.

4. Sampling and laboratory methods

About 90% of the samples were taken using a portable drill, the remainder as block samples. Samples were oriented with magnetic and sun compasses. The difference between the directions by the two methods is less than 5°. Thirty-one samples were taken from five diabase dykes (Fig. 2) and 64 from nine quartz porphyry dykes. In order to carry out baked contact tests, 22 samples were taken from the host migmatites and granites. Unfortunately, full baked contact tests of either diabase or quartz porphyry dykes were not successful owing to too weak intensities and unstable remanences of the host rocks.

In the laboratory, two to five standard 2.5 × 2.2 cm specimens were cut from each sample. All samples were subjected to stepwise alternating field (AF) demagnetization up to 100 mT in steps of 5 or 10 mT using Schonstedt instruments. Thermal demagnetizations were done by Schonstedt furnace in 50° or 100°C intervals below 500°C and in 10° or 20°C intervals up to 680°C. Susceptibilities were measured after each thermal demagnetization step in order to detect mineralogical changes related to heating. AF demagnetization proved to be a better method in isolating different components both in diabases and quartz porphyries. Remanence measurements were done using self-made or JR4 spinner magnetometers.

Principal component analysis (Kirschvink, 1980) was used to calculate the remanence components. Maximum angular deviation (MAD) values of individual linear segments, defined by at least three demagnetization points, were usually 6° and in a few cases 15°. Demagnetization results were examined using orthogonal vector diagrams (Zijderveld, 1967; Torsvik, 1986; Leino, 1991), stereographic projections and intensity decay curves.

Petrophysical properties (NRM intensity, low field susceptibility and density) were measured on each specimen. Saturation isothermal remanent magnetization (SIRM) was induced in a field of 1.4 T for seven samples. Anisotropy of magnetic susceptibility (AMS) was measured from seven diabase samples and from nine quartz porphyry samples using the KLY-2 kappabridge (manufactured by Geofysika Brno). Polished thin sections were studied from at least one sample of each dyke.

5. Palaeomagnetic results

5.1. Quartz porphyries

Quartz porphyries are weakly magnetized, but stable directions were obtained from dykes SA, FA, PN, PV and PS (Fig. 2). In dykes PO, PR, SE and TV detailed AF and thermal demagnetization studies either did not reveal any consistent direction of the NRM at all or only yielded directions corresponding to the Present Earth's Field (PEF) in the sample location. Thus, data from those dykes were rejected. In the quartz porphyries carrying stable magnetizations, three components were typically identified; a low-coercivity PEF, an intermediate-coercivity A_N component and a high-coercivity B component (Fig. 3, Table 1).

In AF-demagnetizations (Fig. 4(a)) the resultant typically moves from steep northerly direction towards a shallow northeasterly direction and then swings back to steep to intermediate northerly direction at high fields. The soft component is removed in low peak fields of the order of 5–10 mT and it is close to the PEF direction. In orthogonal projections the removal of PEF is followed by a curved non-linear trajectory (Fig. 4(c)) illustrating the overlap of PEF and an intermediate-coercivity component A_N. The overlap between the coercivity spectras of PEF and A_N components is, however, partial, leaving windows of non-overlap spectrum at higher fields thus allowing the A_N to be determined. The A_N component has shallow positive or negative inclination and northeasterly declination and it proves to be characteristic of the granite porphyries.

After isolation of the A_N component, the demagnetization trajectories do not pass through the origin indicating the presence of a third, high-coercivity component. This can be seen also in resultant vectors (Fig. 4(a)). In orthogonal vector projections, however, this component, called B, is weakly isolated owing to: (1) slow decay of the demagnetization curve so that no clear orthogonal lines are formed; (2) overlapping of coercivity spectras of A_N and B; (3) decay of the NRM into the noise level of spinner magnetometer before a stable end point is reached. The trajectory in Zijderweld diagrams goes properly towards the origin in only a few samples, while in most samples the data are scattered or offset from the origin. In carrying out the manual line fitting interpretations, two last demagnetization points and the origin were used for some samples to obtain an estimate for the B direction. For some samples the MAD angle used in Kirshvink's method had to be set to 10°–15° for the scattered data.

In an attempt to determine the high-coercivity component, remagnetization circles were used (Halls, 1976). Great circle trajectories of specimens from quartz porphyry dykes PV and PN were plotted on Wulff's net. However, in most cases the great circle trajectories at high coercivities were nearly parallel and short in length so that no well-defined intersection point could be

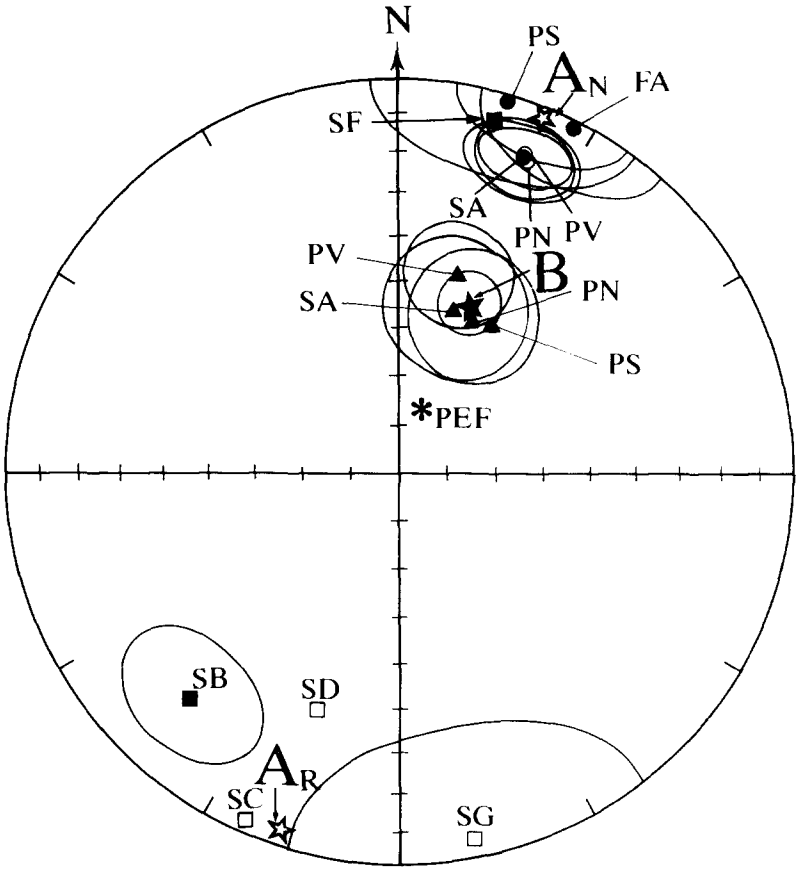


Fig. 3. Dyke mean directions of components A_N (circles) and B (triangles) for the quartz porphyry dykes and of component A_R (squares) for the diabase dykes, with corresponding cones of 95% confidence for those mean directions where the $\alpha 95$-value is less than 30° (see Tables 1 and 2). Component mean directions are shown as stars. SF is the mean direction of the most altered diabase dyke SF (normal polarity). Closed (open) symbols denote downward (upward) directions.

obtained. In cases where the great circles were more scattered and long enough, a common intersection point was defined revealing the A_N but not the B direction.

Many quartz porphyry samples loose their measurable remanence at 500°C in thermal demagnetizations (Fig. 5). Besides, in most thermally demagnetized samples a new magnetic mineral was grown at temperatures above 500°–560°C. That was shown as an increase in the remanence intensity and in the bulk susceptibility (Fig. 5). Therefore it was impossible to isolate any stable component at high temperatures. The growth of a new magnetic mineral could take place by laboratory reformation of the primary magnetite which was hydrothermally altered to silicate minerals as a result of heat and fluids from the rapakivi magma (Törnroos, 1984). However, these alterations can also occur in fresh igneous rocks reflecting the different environment of laboratory heating when compared with natural redox conditions (Tarling, 1983).

In a few cases the remanence intensity remains measurable above 500°C without mineralogical changes. Laboratory unblocking temperatures of magnetization A_N range between 200° and 560°C indicating that titanomagnetite is the carrier of magnetization. Laboratory unblocking temperatures of magnetization B range from 350° to 620°C, indicative of magnetite and haematite. Also the high coercivities (70–100 mT) in AF demagnetization, which did not cause significant decay of the natural remanence, indicate that the carrier of component B is very fine grained magnetite or haematite.

5.2. Diabases

In the diabases, a low-coercivity component with steep positive inclination and northerly declination and a high- coercivity component with shallow to intermediate positive or negative inclination and southerly declination were isolated (Fig. 3, Table 2). The high-coercivity component is antiparallel (reversed) to the A_N component of quartz porphyries and is hereafter called the A_R component. Unfortunately, most of the diabase samples were magnetically unstable and had to be discarded. The samples showed equivalent results only in dykes SB and SG (Fig. 2) although the mean A_R direction between the dykes is quite different. Many individual sample direc-

Table 1
Mean palaeomagnetic data for the Sipoo quartz porphyry dykes (latitude, 60.3°N; longitude, 25.3°E)

Dyke	B/N/n	D (°)	I (°)	$\alpha 95$ (°)	k	Plat. (N°)	Plong. E°)	dp (°)	dm (°)	A95 (°)
Component A_N										
FA	4 */11	27.3	3.1	14.8	39.5	27.6	174.2	7.4	14.8	
SA	5 */13	22.1	15.8	10.1	58.4	35.2	178.2	5.3	10.4	
PV	8 */21	21.9	−14.2	7.9	49.8	20.3	182.1	4.2	8.1	
PN	8 */19	23.0	−16.3	8.6	42.3	19.0	181.2	4.6	8.9	
PS	9 */12	16.6	2.5	20.5	8.2	29.6	186.1	10.3	20.5	
Mean	5 */34/76	22.2	−1.8	13.3	34.2	26.4	180.4	6.6	13.3	(7.4)
Component B										
SA	3 */3	19.9	53.8	14.7	71.8	61.0	169.8	14.3	20.5	
PV	6 */8	17.2	47.0	11.2	37.0	55.8	177.7	9.3	14.4	
PN	4 */4	26.5	54.1	13.4	48.3	59.1	159.6	13.1	18.7	
PS	2 */2	32.8	53.7	–	–	56.3	151.3	–	–	
Mean	4 */15/17	23.9	52.2	6.3	215.8	58.4	164.9	5.9	8.6	(7.6)

Note: B/N/n is the number of dykes/samples/specimens used in the statistical calculations: * denotes the statistical level used in mean calculations; D and I are the mean declination and inclination, respectively; $\alpha 95$ is the radius of the circle of 95% confidence; k is the Fisher's (1953) concentration parameter; Plat and Plong are the palaeolatitude and palaeolongitude for the Virtual Geomagnetic Poles (VGP) of different dykes; dp and dm are the semiaxes of the oval of 95% confidence for each VGP; (A95) is the radius of the circle of 95% confidence of the mean VGP poles. All poles are of normal (N) polarity.

tions are stable in dykes SC and SD, but scatter among samples is high (> 50°). Different branches of dyke SC show different remanence directions. Dyke SD contains large (up to 3–5 cm in diameter) plagioclase phenocrysts resulting in compositional variation presumably affecting the magnetic properties of different samples. Owing to large scatter among diabase dykes, the mean direction and mean pole position cannot be regarded as very reliable. However, these reversed directions have been used in interpretations and will be shown below.

The response of diabases to thermal and AF demagnetization is different. Both methods iso-

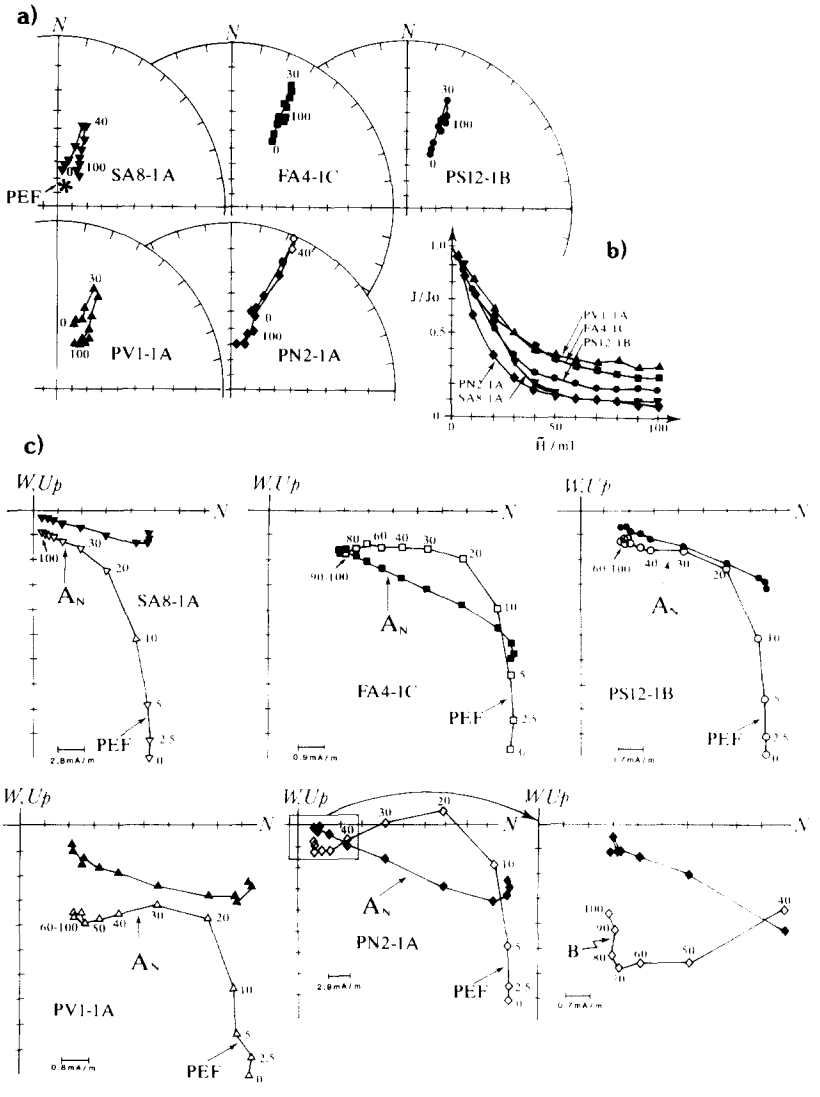


Fig. 4. Examples of AF-demagnetization behaviour typical for different quartz porphyry dykes. (a) Stereographic projections of directional data upon demagnetization, where PEF shows the direction of the present Earth's magnetic field, (b) relative intensity decay curves, (c) orthogonal demagnetization diagrams, where the lower right diagram is an enlargement of the data close to the origin. Open (closed) symbols denote vertical (horizontal) planes. Numbers by each demagnetization step denote peak alternating field (mT).

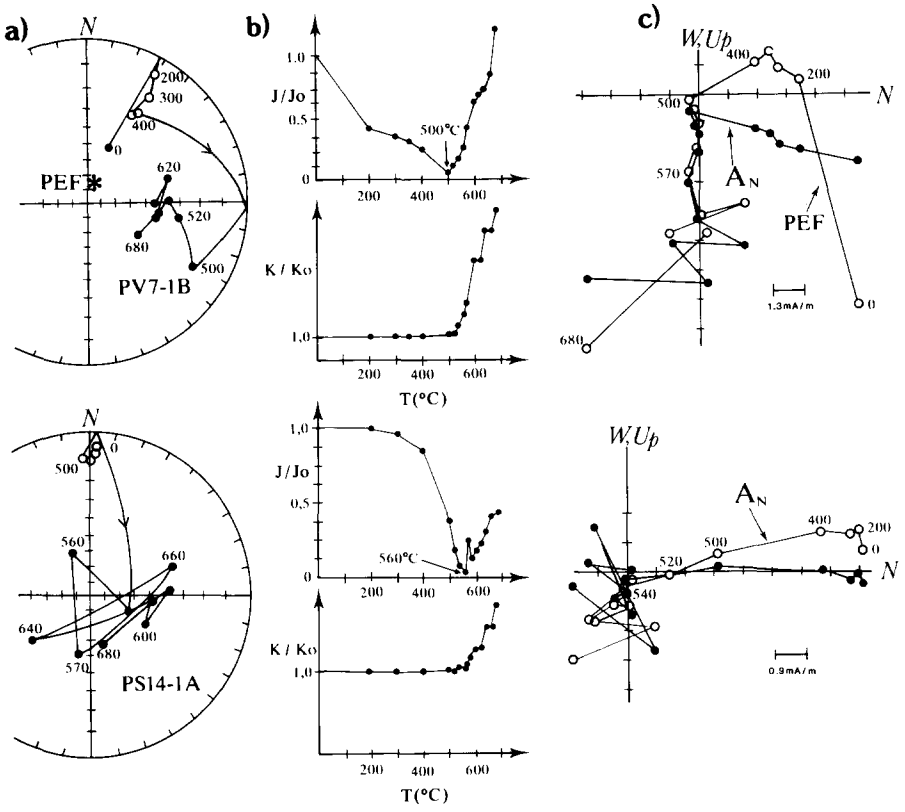


Fig. 5. Examples of thermal demagnetization behaviour typical for quartz porphyries. (a) Stereographic projections, (b) relative intensity decay (upper curve) and behaviour of low field susceptibility upon heating (lower curve), (c) orthogonal demagnetization diagrams. Numbers by each demagnetization step denote temperature (°C). Other symbols as in Fig. 4.

late a northerly steep positive magnetization but AF demagnetization also reveals the shallow southward pointing A_R component. As shown in Fig. 6, the resultant vectors move along a curvilinear trajectory from steep positive northerly inclination to shallow southeasterly or southwesterly inclination, implying the occurrence of two components. The steep positive component is close to the PEF direction and presumably represents a comparatively large viscous remanence (VRM) as

Table 2
Mean palaeomagnetic data for the Sipoo diabase dykes (latitude, 60.3°N; longitude, 25.3°E)

Dyke	B/N/n	*D* (°)	*I* (°)	α95 (°)	*k*	Plat (N°)	Plong (E°)	*dp* (°)	*dm* (°)	A95 (°)	Pol
Component A_R											
SB	8 */13	222.8	23.5	13.9	16.8	9.8	162.9	7.9	14.8		R
SC	4 */ 6	203.5	−4.9	51.8	4.1	29.4	178.1	26.0	51.9		R
SD	3 */ 6	198.6	−36.3	52.4	6.6	47.8	179.0	35.5	61.0		R
SG	3 */ 3	168.7	−7.02	27.8	20.7	32.4	218.7	14.1	28.0		R
SF	3 */ 3	15.5	9.9	52.9	6.5	33.4	186.8	27.0	53.5		N
Mean	4 */18/28	198.0	−6.5	40.1	6.2	31.6	183.6	20.2	40.3	(30.7)	R

Note: See symbols and abbreviations in Table 1. Pol denotes R or N polarity (reversed or normal, respectively). Dyke SF has not been used in the mean calculation.

also indicated by the rapid decay in the first demagnetization steps. The A_R magnetization is more difficult to isolate since the PEF component was not always completely removed owing to overlapping coercivity spectras. In many samples remagnetization circles are generated but stable end points (SEP) shown as linear trajectories in orthogonal plots are reached in only a few samples. In some cases where orthogonal plots did not decay straight to the origin, we have used the SEP direction of the resultant vectors if three or more points in the last demagnetization steps are clustered together. In some samples the superimposed low-coercivity component has no consistent orientation (e.g. specimen SG6-1B, Fig. 6), signifying that the soft component is widely dispersed and of no palaeomagnetic importance.

Thermal demagnetization (Fig. 7) defines only a steep positive magnetization component, which prevails throughout the whole range of blocking temperatures and precludes identification of any underlying components. Owing to the steep inclinations (about 90°) the declinations of different samples are widely scattered. The remanence direction corresponding to the steep component, isolated by AF demagnetization is interpreted as PEF, although it is resistant to heating and cannot be removed even at 580°C. This may be due to large overlap of the unblocking temperatures for multidomain (MD) grains, carrying the soft

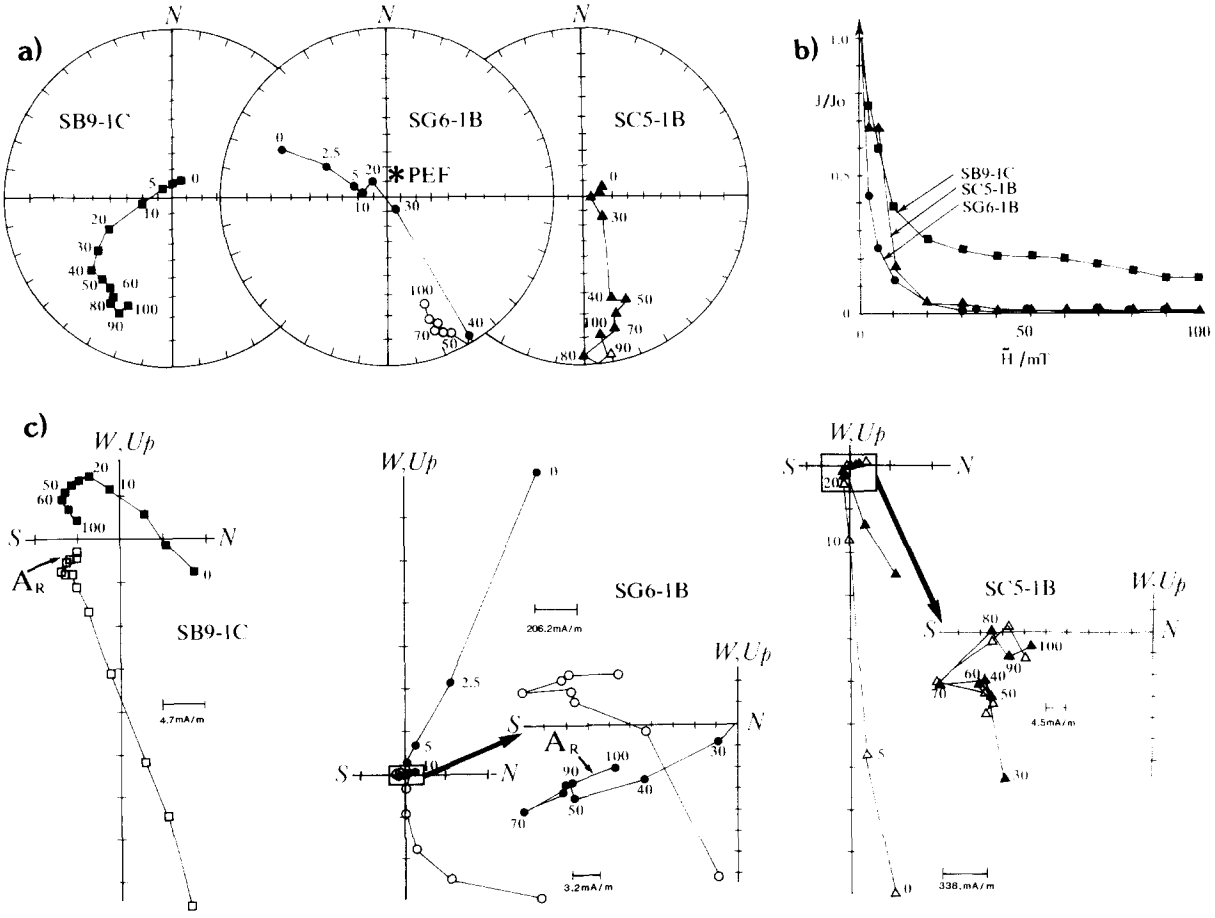


Fig. 6. Examples of AF demagnetization behaviour typical for diabase dykes. (a) Stereographic projections, (b) relative intensity decay, (c) orthogonal demagnetization diagrams. Data points close to the origin are shown enlarged for specimens SG6-1B and SC5-1B. Other symbols as in Fig. 4.

viscous PEF magnetization and single domain (SD) grains carrying the primary thermoremanence. Because the viscous component has been acquired at low temperatures it is possible to demagnetize it by low alternating fields (see e.g. McClelland-Brown, 1982).

Thermal laboratory unblocking temperatures at about 580°C indicate that magnetite is the main magnetic carrier. Above 580°C the remanence directions become widely scattered despite the intensities being above the noise level. This observation suggests that there is no haematite present. A small change at about 350° in the demagnetization decay curve may indicate that small amounts of pyrrhotite (Curie temperature of 325°) are present.

Dyke SF is the most altered diabase dyke compared with other dykes in the area. In four out of five samples, the remanence has a shallow northerly direction which is antiparallel to the high-coercivity A_R direction in other diabases. It corresponds to the A_N direction of the quartz porphyries. In most specimens the resultant direction moves from PEF towards A_N, but does not reach the stable end point. However, the stable end point is reached in specimen SF3-1A (Fig. 8) and is shown in orthogonal vector plot (Fig. 8(c)).

A high-coercivity A_N component is also isolated in a few samples of the best preserved diabase dyke SB which carries the most stable remanence direction A_R (see Table 2 and Fig. 9). The end point (A_N) is not due to erratic behaviour in the last steps of demagnetization showing considerable intensity increase, since the scatter is within the measurement precision. Moreover, the same high field A_N direction has been observed in different samples. Because its occur-

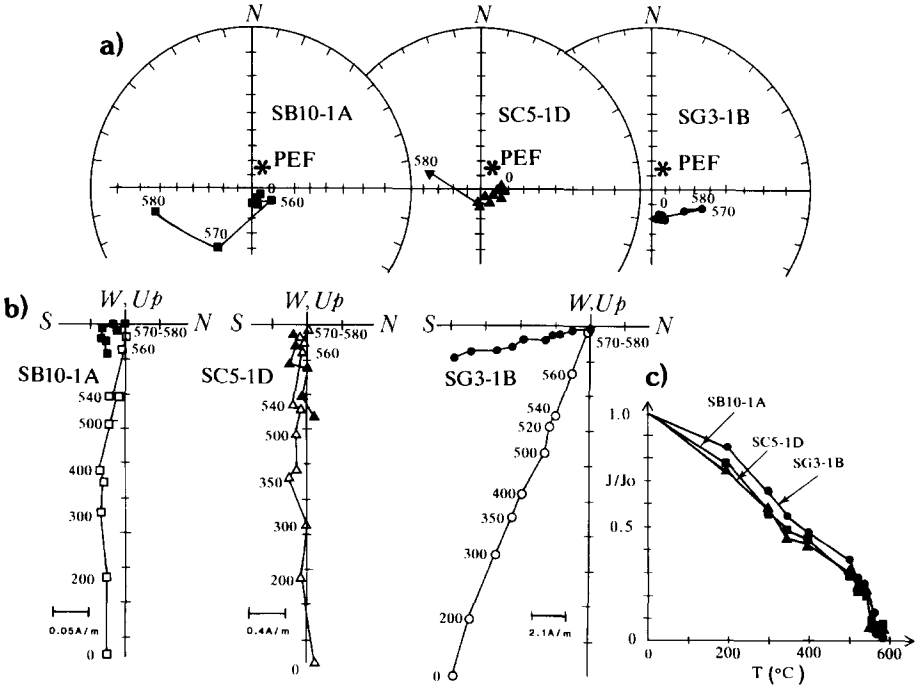


Fig. 7. Examples of thermal demagnetization behaviour typical for diabases. (a) Stereographic projections, (b) orthogonal demagnetization diagrams, (c) relative intensity decay curves. Other symbols as in Fig. 4.

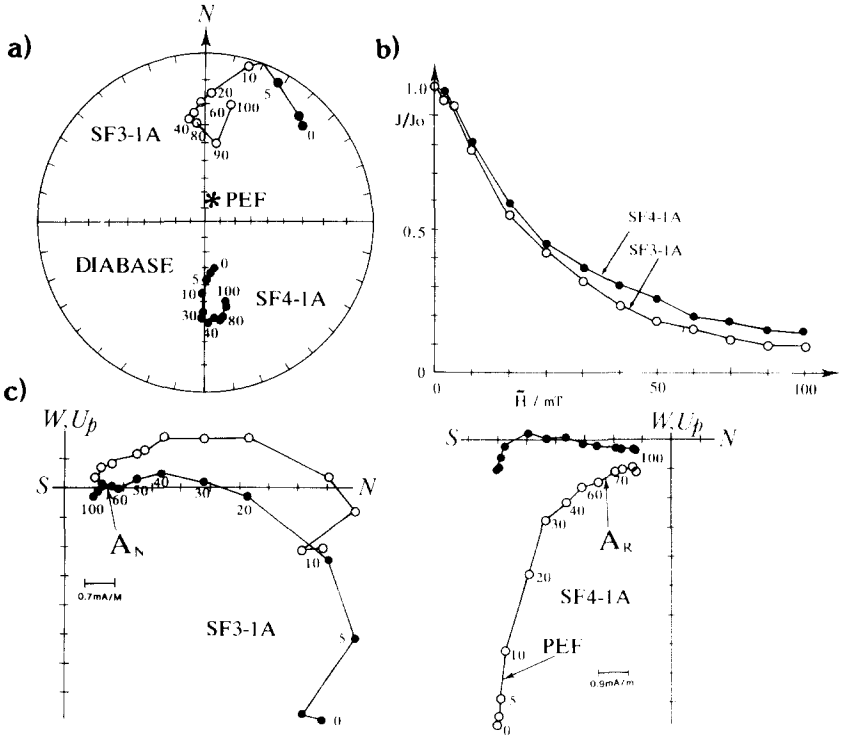


Fig. 8. Examples of AF-demagnetization behaviour for specimens of the most altered diabase dyke SF. Specimen SF4-1A carries the reversed polarity remanence, while specimen SF3-1A is completely remagnetized in normal polarity direction. Other symbols as in Fig. 4.

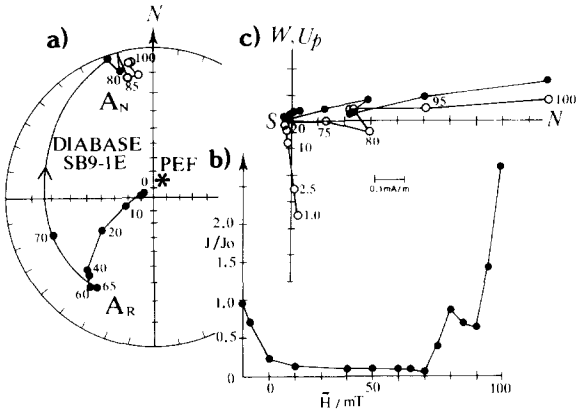


Fig. 9. AF-demagnetization behaviour of specimen SB9-1E from the best preserved dyke SB. The specimen shows the high-coercivity component A_N as a stable end point (a), as an increase of relative intensity (b) and as a straight line (c). The characteristic component A_R for diabases is isolated in a narrow coercivity range, 50–70 mT. Other symbols as in Fig. 4.

rence is limited to only a few samples, it has not been listed in Table 2. The origin of this magnetization will be discussed later.

6. Petrography

6.1. Quartz porphyries

Thermal demagnetizations of the samples, which displayed no mineralogical alterations upon heating, showed that titanomagnetite is the main remanence carrier in the quartz porphyries (T_{lub}s between 550° and 580°C). Petrographic studies show that titanomagnetite has several forms of occurrences. Some dykes contain large euhedral magnetite phenocrysts with ilmenite lamellae, showing that deuteric oxidation at high tempera-

tures has taken place. The fine-grained magnetite in the groundmass is homogeneous. In hydrothermal alteration magnetite or ilmeno-magnetite is partly altered to form amphibole, biotite and calcite in reaction with orthoclase or plagioclase. Hydrothermal alteration can also be seen as corroded titanomagnetite grains. Skeletal pseudomorphs built up of amphibole and biotite have in some cases totally replaced the oxides. In addition, magnetite occurs as very small laths in spherulites. The grain size distribution of magnetite is strongly bimodal but the relative abundance of fine and coarse grains is not consistent from sample to sample. This may be an explanation of the variation of PEF contamination in the samples.

AF demagnetization (e.g. Fig. 4) showed the occurrence of a very hard, high-coercivity remanence B which can reside in haematite or in very fine grained magnetite. However, thin sections have not revealed haematite in the groundmass or in the spherulites. The limit of optical resolution is about 1 mm and evidently the hard remanence resides in submicroscopic grains.

The phenocrysts in which the effective magnetic grain size is small owing to ilmenite lamellae (Haggerty, 1976) and the small magnetite grains in the groundmass are probable carriers of the primary remanence. Large magnetite grains which dominate in most samples are responsible for the large viscous magnetizations (PEF).

6.2. Diabases

Diabases show subophitic texture with lath-shaped plagioclase. Some samples from the contacts show a clear flow structure. Most diabases contain primary augite which is often altered to biotite, amphibole and magnetite. Magnetite also occurs as large anhedral grains, as corroded grains and as fine-grained aggregates. Dyke SF is the most altered diabase dyke. Pyroxenes have totally been replaced by amphibole, biotite and magnetite. Plagioclase is also strongly sericitized in dyke SF. Magnetite occurs as larger aggregates consisting of small grains and as pure, lath-shaped euhedral grains. These magnetite grains are probably crystallized during later events (see discussion later).

In the diabases the large magnetite grains are dominant and are probably the carriers of the large viscous magnetization (PEF). Opaque grains carrying the primary remanence are difficult to ascertain, but based on the relatively hard coercivities to which component A_R persists, the carrier can also be submicroscopic magnetite grains. Murthy et al. (1981) have shown that duality in magnetic grain sizes can produce controversial magnetic properties, like multidomain behaviour in rocks that carry stable remanence.

7. Bulk petrophysics

7.1. Quartz porphyries

The quartz porphyries are generally weakly magnetized (Table 3, Fig. 10). Susceptibilities vary from 400 to $10\,250 \times 10^{-6}$ SI, with a mean of 4670×10^{-6} SI. Remanence intensities vary from 3 to 300 mA m^{-1}, with a mean of 111 mA m^{-1}. The wide range of susceptibility and remanence intensity values is probably due to variation in grain sizes. Koenigsberger's Q values are between 0.25 and 2.5. The magnetic properties vary to some extent in different dykes, although based on those differences the dykes cannot be separated from each other. Most likely the dykes belong to the same dyke swarm.

Dykes PS, PR and SE carry the weakest magnetization. Stable remanences are found in dyke PS while samples from dykes PR and SE are too weakly magnetized to yield reliable palaeomagnetic results. Dykes FA, SA, PV and PN, carrying moderate magnetization intensities, have stable remanences which can be used in palaeomagnetic interpretations. Samples from these dykes have the highest Q-values further indicating the stability of remanence. These results are in agreement with petrographic observations that the matrix magnetite is fine grained and the magnetite phenocrysts have small effective grain size owing to ilmenite exolution which increases the intensity and stability of the remanences.

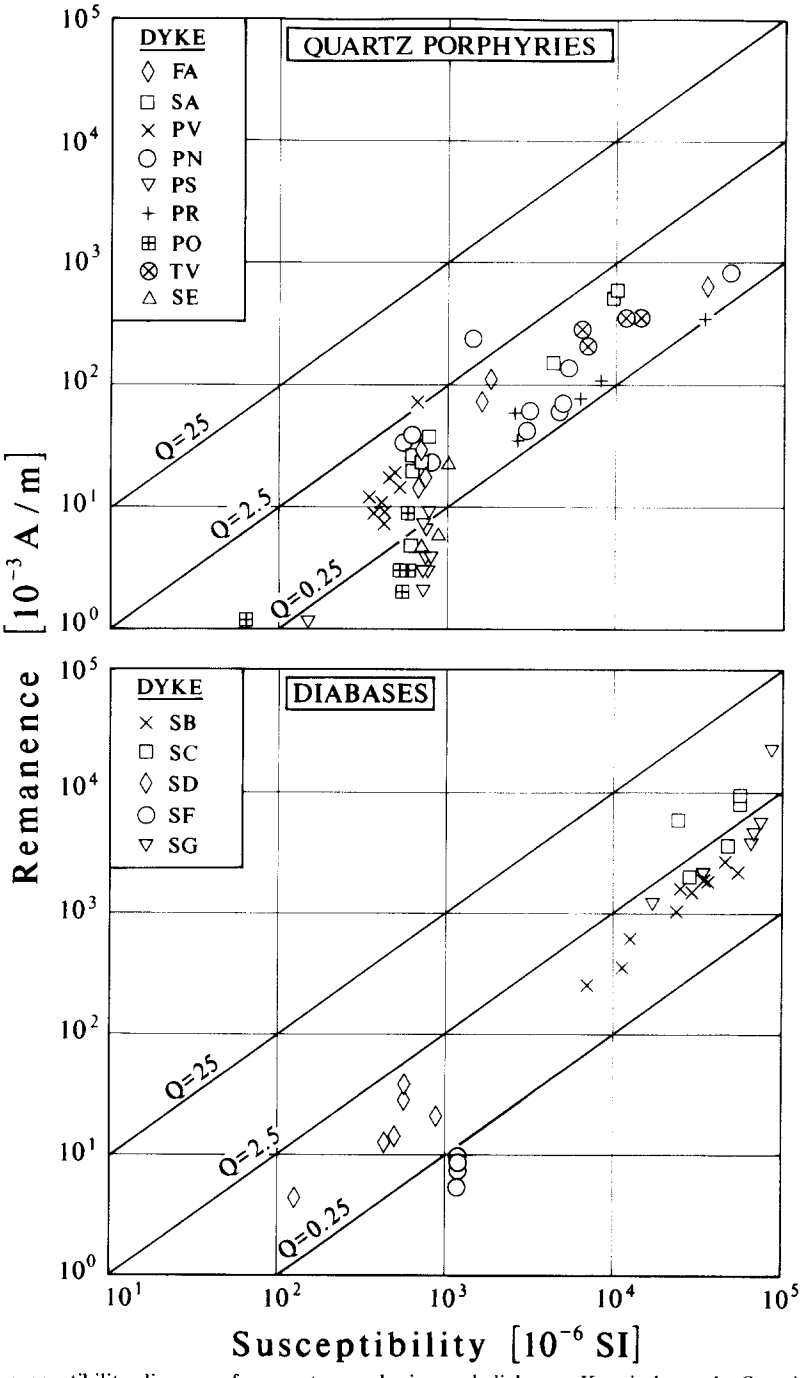


Fig. 10. Remanence–susceptibility diagrams for quartz porphyries and diabases. Koenigsberger's Q ratios are shown as line Palaeomagnetic results for quartz porphyries were obtained from dykes PV, PN, PS, FA and SA.

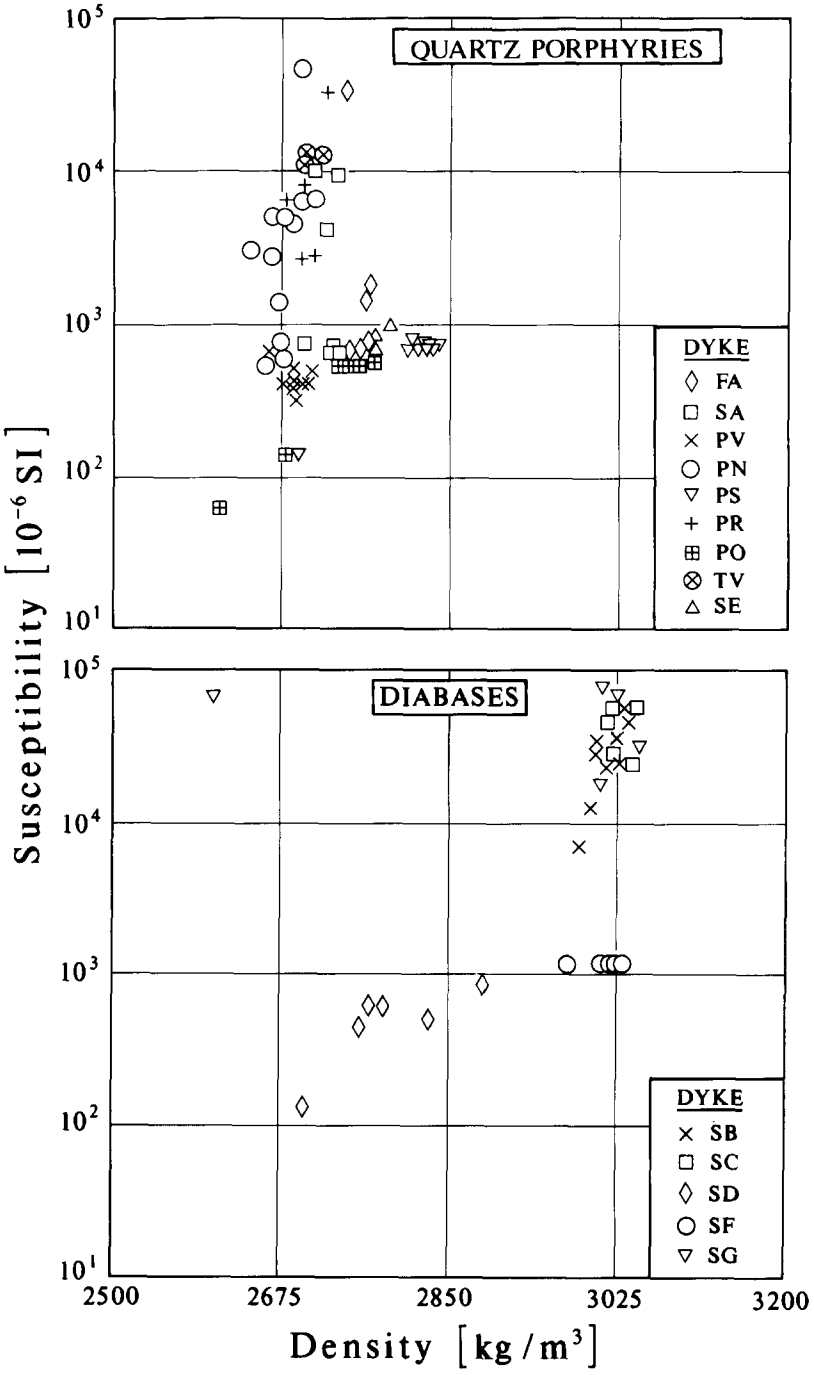


Fig. 11. Susceptibility–density diagrams for quartz porphyry and diabase dykes.

Densities (Table 3, Fig. 11) range between 2670 and 2810 kg m^{-3}, the mean value being 2728 kg m^{-3}. The densities and susceptibilities of dykes FA and SA are similar. In dykes PV and PN the densities are similar although the magnetization is variable. Dyke PS shows the highest

density values, but low susceptibility and magnetization values which indicate that the high densities are mainly due to heavy silicate minerals.

7.2. Diabases

Diabase dykes SB, SC and SG show the highest magnetization values (Table 3, Fig. 10). Susceptibilities of the diabases range from 12 700 to 73 700 $\times 10^{-6}$ SI and the remanence intensities from 600 to 9000 mA m^{-1}. The Koenigsberger's Q ratios vary from 0.1 to 6.8. Unlike the quartz porphyries, the diabases show a void of midrange susceptibilities. The high values obviously represent typical values of diabases but the low ones are due to exceptional properties of dykes SD and SF. In dyke SD the low magnetization values are due to extremely large (diameter up to 5 cm) plagioclase grains that occupy most of the specimen and thus decrease the relative proportion of iron oxide minerals. In dyke SF the decreased susceptibility and remanence intensities are most probably due to hydrothermal alteration that has not taken place in other dykes. The alteration was also shown by the absence of primary pyroxenes at the expense of newly crystallized amphibole and biotite. The decrease of susceptibility can be due to reduction of magnetic grains and formation of less-magnetic and non-magnetic minerals (Lapointe et al., 1986). Dyke SF is the only diabase which carries component A_N.

In diabases the densities range between 2790 and 3030 kg m^{-3}, the average value is 2960 kg m^{-3}. The density of dyke is SF equal to the density of other diabases while in dyke SD the densities are clearly lower than in the other diabases indicating a higher amount of plagioclase crystals.

8. Anisotropy of magnetic susceptibility

In palaeomagnetic studies it is essential that the remanence directions reflect the Earth's ambient magnetic field direction. The NRM direction can deviate from the ambient field direction for three reasons: (1) owing to tectonic tilting of the dykes; (2) owing to deflection of the rema-

Table 3
Petrophysical properties of the Sipoo dykes

Dyke	B/N	D	K	J	Q	Pol
Quartz porphyries						
FA	6	2759	6522	146.6	0.56	N
SA	10	2726	5935	239.0	1.01	N
PV	9	2689	447	19.2	1.08	N
PN	10	2672	7043	152.2	0.54	N
PS	12	2811	668	3.9	0.15	N
PR	6	2716	404	3.0	0.19	
PO	5	2701	10250	120.4	0.30	
TV	5	2706	9903	304.0	0.77	
SE	3	2776	852	11.8	0.35	
Mean	9/66	2728	4670	111.1	0.55	
Diabases						
SB	9	3016	31005	1587.7	1.29	R
SC	5	3029	42565	5797.2	3.42	R
SD	6	2787	508	18.4	0.91	R
SF	5	3013	1180	7.9	0.17	N
SG	6	2954	56120	6685.8	2.99	R
Mean	5/31	2960	26276	2818.4	1.76	

Note: B/N is the number of dykes/samples used in the statistical calculations; D is density (kg m^{-3}); K is mean susceptibility (10^{-6} SI); J is remanence intensity (mA m^{-1}); $Q = J/(K \times H)$ where H is the Earth's magnetic field intensity; Pol is the palaeomagnetic polarity, where determined.

nence towards the plane of the dyke (Evans, 1987); or (3) owing to magnetic anisotropy. First, in the Sipoo dykes the tectonic tilting is probably insignificant because the dykes are subvertical and represent the youngest, undeformed formations in the area. Second, there is no correlation between the strike of the dykes and the declination of the primary remanence component *A*. This was verified by increased scatter of remanence directions after the dyke strike directions were reduced from the *A* component declinations. Third, as will be shown below, the deflection of remanence directions from the ambient field owing to the anisotropy of magnetic susceptibility (AMS) has only minimal influence on the palaeomagnetic results.

AMS was measured from a few samples of the dykes carrying stable remanences. According to Stephenson et al. (1986) the NRM is not significantly affected if the ratio between maximum and minimum susceptibilities (degree of anisotropy) is lower than 5%. However, a weak anisotropy of susceptibility does not rule out the possibility that NRMs are deflected, because the anisotropy of TRM can lead to much larger deviations of the NRM from the ambient field direction (Stephenson et al., 1986; Tarling and Hrouda, 1993). Here we only report AMS data which were measured with the susceptibility bridge.

The degree of anisotropy in diabases is so low (4–8%, Table 4) that it has not markedly deflected the characteristic remanence directions. All measured diabase samples are from the margins of the dykes. In samples of quartz porphyries, taken from the interior parts of the dykes, the degree of anisotropy ranges from 1 to 9% which is comparable with the anisotropy of diabases. However, in those two quartz porphyry samples (FA5-1B and PN8-1E) which were taken from the margins of the dykes, the degree of anisotropy is higher, 20% and 25%, respectively. This anisotropy is probably due to large magnetite grains aligned along the margin of the dykes. Törnroos (1984) has shown that near the contacts, biotite and hornblende become elongated indicating a clear flow structure, probably formed during the emplacement of the dykes. However, the remanence directions in the margin and the interior samples do not differ significantly. Thus, the remanence was blocked after this flow structure was formed. Therefore it is probable that the AMS is carried by other grains than the remanence. In many studies (e.g. Stephenson et al., 1986; Hyodo and Dunlop, 1993) it has been shown that large multidomain grains (MD) which are the carriers of soft VRM are also the main source of the AMS, while small single-

Table 4
The degree of anisotropy (*P*) in Sipoo dykes

Quartz porphyries		Diabases	
Specimen	*P*	Specimen	*P*
PV4-1B	1.09	SB1-1D	1.06
PV7-1E	1.08	SB4-1D	1.08
PV9-1F	1.07	SC2-1D	1.07
PV10-1E	1.05	SC5-1E	1.06
PN8-1E	1.25	SG2-1B	1.08
PS7-1E	1.03	SG4-1D	1.04
PS10-1F	1.01	SF3-1E	1.01
FA5-1B	1.20		
FA6-1D	1.02		
Mean	1.09	Mean	1.06

Note: *P* is the ratio between maximum and minimum susceptibility.

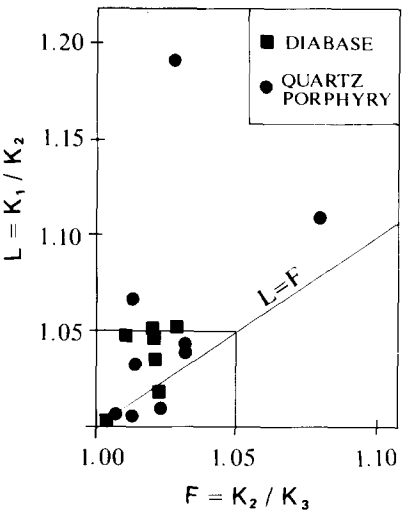


Fig. 12. Magnetic foliation (F)–lineation (L) diagram for quartz porphyries and diabases. The rectangle in the left corner characterizes magmatic flow fabrics (see Hrouda and Chlupacova, 1980). *K*1, *K*2, and *K*3 are the maximum, intermediate and minimum susceptibilities, respectively.

domain (SD) grains carrying stable remanences affect the magnetic anisotropy less.

In the magnetic anisotropy plot (Fig. 12) most samples from both the diabases and quartz porphyries are plotted on the area characteristic for magmatic flow fabrics (Hrouda and Chlupacova, 1980). AMS can indicate the flow direction, since especially for mafic dykes it is typical (e.g. Knight and Walker, 1988) that the minimum susceptibilities ($K3$) are perpendicular to the dyke plane and the maximum susceptibilities ($K1$) are aligned in the direction of magma flow. Both in diabases and quartz porphyries (Fig. 13) there is a tendency for $K3$-axes to be perpendicular to the average dyke plane and for the $K1$-axes to be within the dyke plane. In quartz porphyries the inclinations of $K1$-axes are shallow which indicates a shallow trajectory of magma flow. The two quartz porphyry samples from the margins of the dykes are dominated by magnetic lineation (Fig. 12) and the $K3$-axes are perpendicular to the dyke plane while the $K1$-axes have intermediate inclinations within the dyke plane. In diabases the inclinations of maximum axes vary between different specimens. Owing to the small number of specimens in AMS measurements, the conclusions are somewhat uncertain.

9. Rock magnetic studies

Modified Lowrie–Fuller (Lowrie and Fuller, 1971; Dunlop, 1983) tests were carried out to one quartz porphyry and one diabase sample (Fig. 14) in order to study the magnetic grain sizes and thus the ability of rocks to carry stable remanences. The following procedure was used.

(1) The NRM of the specimens was demagnetized by AF in steps of 0, 2.5, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 mT.

(2) Anhysteretic remanent magnetization (ARM) was induced in 100 mT peak AF and 50 mT and 75 mT direct fields (DC), respectively.

(3) ARMs were AF demagnetized in the same steps as in (1).

(4) SIRM was induced in the field of 1.4 T.

(5) SIRM was AF demagnetized in the same steps as in (1).

In a modified Lowrie–Fuller test the normalized AF demagnetization curves of NRM, ARM and IRM are compared. ARM is used, since it has similar coercive force spectrum to thermal remanent magnetization (TRM) and by using ARM possible mineralogical changes during heating can be avoided. The basic principle in the modified Lowrie–Fuller test is that SIRM is less

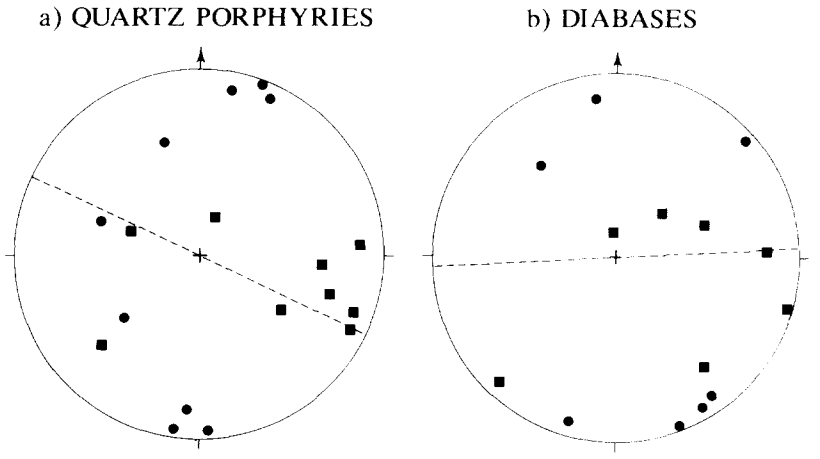


Fig. 13. Directions of minimum (dots) and maximum (squares) susceptibility for specimens of (a) quartz porphyries and (b) diabases, plotted on the lower hemisphere of equal-area stereonets. Average strikes of dykes are given by dashed lines.

stable than ARM in samples exhibiting an abundance of SD and/or PSD magnetite grains, while the opposite is true for MD behaviour.

Quartz porphyry and diabase samples used in the Lowrie–Fuller test show contrasting behaviour with each other. Quartz porphyry sample PN8-1F (Fig. 14(a)) shows a typical SD behaviour at lower fields where SIRM is less stable than ARM and NRM which are very similar. There is, however, a cross-over from a SD- to a MD-type configuration around 40 mT. As Dunlop (1983) has pointed out, this kind of behaviour, in addition to closely spaced curves, indicates that there is a mixture of coarser MD titanomagnetite grains and finer grains of SD or PSD size. In this sample the result agrees with the directional data which show a small contribution of viscous PEF component, probably carried by MD grains, but a large contribution of intermediate- to high-coercivity A_N and B components which are carried by SD/PSD grains.

In the diabase sample SB7-1A (Fig. 14(b)),

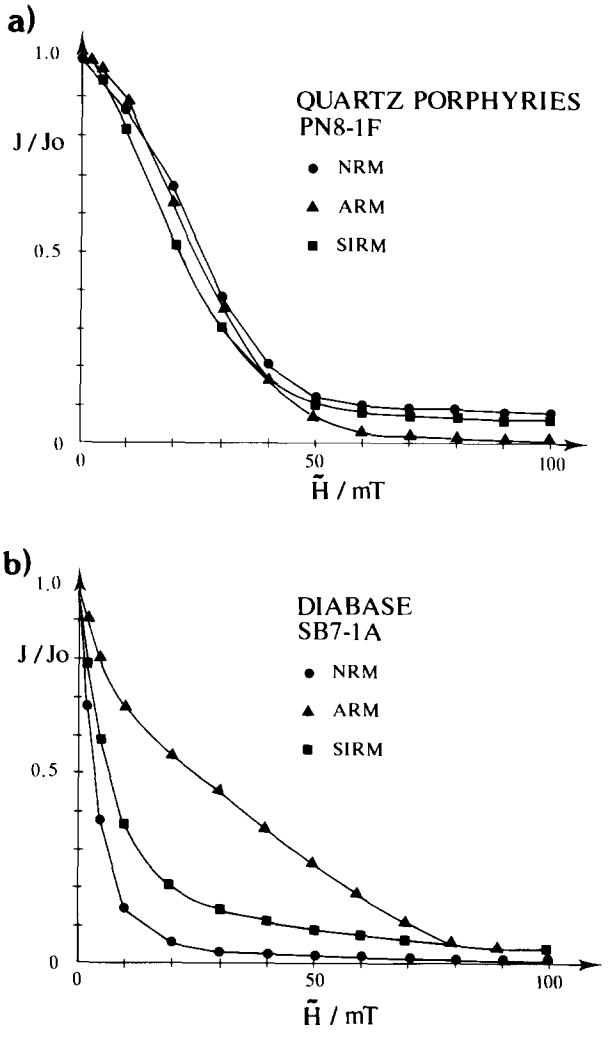


Fig. 14. Lowrie–Fuller test showing the NRM, ARM and SIRM curves for quartz porphyry (a) and diabase (b) samples. The ARM produced using two DC fields (50 mT and 75 mT) showed no difference in AF behaviour.

SIRM is less stable than ARM which indicates SD/PSD magnetic grain size. However, the results are bimodal; SIRM and ARM curves are widely separated; SIRM is softer than ARM, but ARM has sharp initial decay which is followed by slow decay at high coercivities. This behaviour shows dominance of MD fractions and a smaller contribution of the SD fraction to the ARM demagnetization curve. These results are consistent with petrographic and NRM results which show that a significant proportion of large MD magnetite grains carry a viscous component (VRM), indicated by a notable decrease in intensity during the first steps in the AF-demagnetization of NRM, while a hard remanence (component A_R) is carried by small magnetite grains, probably not visible in microscope studies. Examples of the IRM acquisition curves are shown in Fig. 15. At low magnetic induction values, the curves for quartz porphyries exhibit a gradual increase, suggesting that the remanence carriers are mainly pseudo-single and SD magnetite with a small contribution of haematite. The slightly steeper gradient of the IRM curve of the diabase below 0.05 T suggests that the carrier is magnetite, although according to Lowrie (1990), maghemite, pyrrhotite and multidomain haematite coercivities also fall in this range. The observation that all samples reached saturation at 100 mT further indicates that magnetite is the remanence carrier.

10. Discussion

10.1. Origin of magnetizations

10.1.1. Quartz porphyries

Magnetization A_N has been isolated in quartz porphyries containing primary magnetite grains with exolution ilmenite lamellae. The equilibration of primary titanomagnetite takes place between 750° and 550°C (Haggerty, 1976) which corresponds to the temperatures (700°–530°C) at which the phenocrysts of the Sipoo quartz porphyries have solidified (Törnroos, 1984). After the emplacement of the quartz porphyries the dykes have been hydrothermally altered to some degree as a consequence of the ascent of the Onas rapakivi magma, which has also altered the magnetite in the dykes. However, as Bol et al. (1992) have observed, hydrothermal alteration does not affect the Ti-poor magnetite with ilmenite lamellae, but the high-temperature oxidation of titanomagnetite is preserved from the time of primary cooling of the rock. Hence, based on petrographic evidence, it is suggested that component A_N can be a primary thermoremanence (TRM) formed during the crystallization of phenocrysts. In dykes without magnetite phenocrysts, the direction of the A_N magnetization corresponds to the remanence direction in dykes

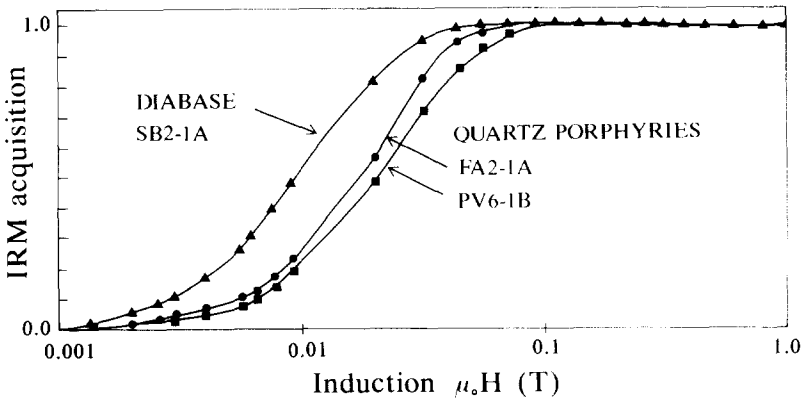


Fig. 15. Normalized IRM acquisition curves for quartz porphyry specimens FA2-1A and PV6-1B and for diabase SB2-1A.

where phenocrysts occur. Thus, the fine-grained groundmass carries the same magnetization as the phenocrysts.

The origin of remanence in dykes can be studied by performing a baked contact test (see e.g. Everitt and Clegg, 1962; Buchan and Halls, 1991). In this test the remanence in the dyke is considered primary, if the dyke and baked host rock carry a consistent, stable direction of magnetization, while unbaked host rock has a stable but different remanence direction. In Sipoo quartz porphyries, unfortunately, baked contact tests could not be carried out owing to the weak and unstable remanent magnetizations in the host granites and migmatites.

In another method (Halls, 1986) the remanence is considered primary if it remains constant along a dyke but varies between dykes. Differences of remanence directions between dykes are due to the short time interval in which the dykes cool and record secular variation of the Earth's magnetic field. Based on the orientation, geographical location and similar petrophysical properties, dykes FA and SA may represent exposures of the same dyke. The same is true for dykes PV and PN. Fig. 3 shows that the A_N magnetization in dykes FA and SA and in dykes PV and PN are similar but the magnetization of FA/SA differs significantly from that of PV/PN. This difference probably reflects secular variation of the Earth's magnetic field and hence support the primary origin for component A_N. However, the incomplete isolation of the PEF component can also contribute to the scatter of NRM directions.

The origin of the high-coercivity B component in the quartz porphyries is more difficult to interpret. As has been shown in many palaeomagnetic studies (e.g. Buchan, 1978), a hard magnetization is not necessarily the oldest one, especially in cases where the relative ages cannot be established by unblocking temperatures. High coercivities to which B persists appear, however, incompatible with B being a true recent viscous overprint. According to Törnroos (1984), hydrothermal fluid from the rising Onas rapakivi magma is the only event that has altered the quartz porphyries since their emplacement. The age difference between the rapakivi granites and quartz porphyries is so small that palaeomagnetic directions of the primary thermoremanence of the dykes (component A_N) and remanence formed during the subsequent hydrothermal alteration can hardly be separated. Hence, the direction of B is clearly different from that of A_N and cannot have been acquired in this alteration process. Subsequently, there may be other processes which cannot be detected by petrographic observations, but are seen palaeomagnetically. In the next we suggest that B is a chemical remanence (CRM) acquired at temperatures much below the maximum blocking temperature in the presence of a magnetic field. Three alternative processes in which B may have been acquired are proposed.

First, B could be a chemical remanence acquired as a result of groundwater circulation at some time during the geological history of the dykes. Dobson and Heller (1992) have found that groundwater circulation in dykes may cause oxidation of pyrite to magnetite and formation of secondary haematite. The heat comes from warm saline groundwater. In Sipoo quartz porphyries, haematite was not observed in thin sections and hysteresis properties did not indicate the existence of haematite either, although the high coercivities during AF demagnetization and T_{lub}s up to 620°C may point to haematite. However, since the intensity of saturation magnetization of magnetite is about 250 times that of haematite small amounts of haematite may go unnoticed in IRM measurements. Hydrothermal groundwater is a possible mechanism for producing chemical remanence in Sipoo area, where no later processes with elevated temperatures have been reported. This assumption, however, requires more supporting evidence from the geology of the area as a whole.

In the second explanation B may be a composite of the PEF and Subjotnian directions, with no geological meaning. In many recent studies it has been cautioned that a chemical remanence possibly does not reflect the past ambient geomagnetic field, but may be the result of many factors (e.g. Heider and Dunlop, 1987; Pick and Tauxe, 1991; Kelso et al., 1991; Walderhaug et al., 1991; Walderhaug, 1992). If a CRM has been acquired during alteration of magnetite to haematite, or

during formation of new magnetite, it has formed under the influence of both the external magnetic field and the initial remanence. Walderhaug (1992) has pointed out that the newly acquired chemical magnetization can be either parallel with the ambient field or it can be a stable direction between the prealteration remanence component and that of the ambient field.

In Sipoo quartz porphyries the relative intensity of PEF, A and B components varies in different samples. Moreover, the directions of the three components all settle on the same great circle, where B represents the intermediate direction between PEF and A. Hence, it is possible that the primary stable A_N component has magnetostatically biased the magnetization direction from PEF towards A_N and produced an apparent intermediate component B. Most specimens carrying the B remanence represent the uppermost, near-surface parts of the cores and hence it is possible that some magnetite has oxidized to haematite, which is able to carry a very hard weathering-type remanence.

Because the diabase dykes are about the same age as the quartz porphyry dykes and have undergone the same tectonic processes, we would expect to find the B component in the diabases, too. However, in the diabase samples we have not found the B component. This may be an indication that component B indeed is a composite of PEF and A_N in the quartz porphyries which were chemically more vulnerable to oxidization and remagnetization by weathering.

Finally, the third explanation for the origin of the B component assumes that there has been a younger geological remagnetization event merged with somewhat elevated temperatures which has not been found by other methods, but is detected palaeomagnetically as the B overprint. This alternative will be discussed later.

10.1.2. Diabases

As discussed before, the baked contact test is the most efficient method to distinguish between primary and secondary remanences. For the diabases, a full baked contact test could not be performed, because the host rocks are too weakly magnetized and carry unstable remanence. However, the characteristic A_R magnetization of diabases was isolated (Fig. 16) in one Svecofennian granite sample, about 1 m from the contact of the diabase dyke SF (width of the dyke at this site is 3 m). This magnetization probably represents a partial thermoremanent magnetization (pTRM) acquired during the dyke emplacement, although A_N remanence dominates in the dyke itself. Hence, the reheating took place during the reversed polarity time and the baked granite has retained this direction and has not been remagnetized later as have the diabases (see below). Svecofennian rocks typically have a northwesterly intermediate downward direction (e.g. Pesonen et al., 1991) and thus the magnetization isolated at this site differs clearly from it. These data, although based on only one sample from the host

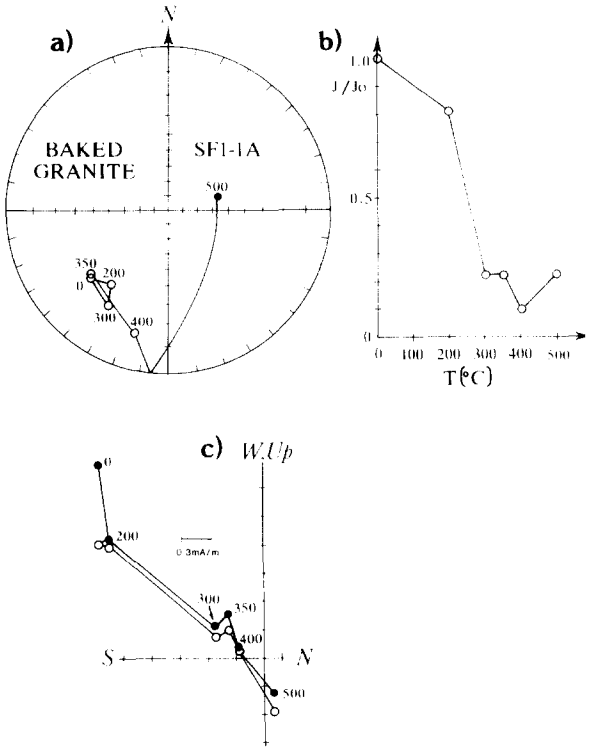


Fig. 16. Specimen SF1-1A from baked Svecofennian granite, about 1 m from the contact of the diabase SF (width of the dyke is 3 m), shows a low-temperature component corresponding to component A_R in diabases. Above 400°C the data become erroneous owing to weak intensities.

rock, suggest that A_R is the primary magnetization of the diabases.

The reversed A_R component occurs in all diabase dykes, but the directions are scattered between samples and between dykes. Significant overlap of the low-coercivity PEF component and high-coercivity A_R components could explain the large scatter. In such a case, however, the directions would be streaked from PEF to A_R, depending on the relative amounts of the components. This is not observed (Fig. 3) but, rather, the directions follow nearly Fisherian distribution. In diabases the scatter between dykes is mostly due to differences in declinations which are best interpreted by rotation about a vertical axis. If the deviation is due to vertical axis rotation, it must have taken place before the emplacement of the quartz porphyries, where corresponding deviations in declinations have not been observed.

The main reason for the scattered data of diabases is presumably related to the hydrothermal alteration resulting from the emplacement of quartz porphyries and Onas rapakivi granites. Dyke SF, which is the most altered diabase dyke, carries magnetization A_N as a secondary component while in the quartz porphyries A_N is primary. Therefore it is suggested that the remagnetization of dyke SF has taken place as a consequence of hydrothermal fluids released from the quartz porphyries and rapakivi granites. In most diabase dykes, the overprint has not been strong enough to entirely remagnetize the primary A_R remanence, but it has increased the scatter. Since the narrow (up to 3 m) Sipoo diabase dykes have cooled on time scales shorter than that of geomagnetic field reversal (see e.g. Hyodo and Dunlop, 1993), the dykes of similar age should have a single polarity. The fact that we have found both polarities in dyke SF also demonstrates that the other one must be secondary. Furthermore, dyke SB, which is the best preserved diabase dyke, carries also the normal polarity magnetization A_N in high-coercivity fields, in addition to a strong characteristic reversed magnetization A_R. The high-coercivity indicates that the magnetization is carried by small, magnetically hard grains which can be either magnetite or haematite. Hence, both in dyke SF and dyke SB the remanence must be a thermochemical overprint acquired during the emplacement of the rapakivi granite and quartz porphyry dykes carrying the same remanent magnetization. This observation also confirms that the diabases are slightly older than the quartz porphyries.

The variable degrees of alteration and consequently the varying proportions of overprinting in the diabase dykes, could be related to the distance of the dykes from the Onas rapakivi granite. Dyke SF is closest to the rapakivi granite of those dykes where consistent palaeomagnetic results were obtained. Distance of dyke SF from the rapakivi is about 12 km while distance from dyke SC, where A_N was not found, is about 20 km (Fig. 2). Thus the extent of remagnetization by hydrothermal fluids is about 12–20 km. Dyke SD which is closest (about 5 km) to the rapakivi granite failed to yield consistent results, but the polarity is certainly reversed. Hence, this dyke shows that hydrothermal fluids from the rapakivi granite have not completely remagnetized the diabases even as close as 5 km from the contact. This indicates that the remagnetization is a very local and selective event and suggests that the hydrothermal fluids proceed along fractures in the bedrock. The fracture density most likely varies in the bedrock producing preferred routes for fluids and consequently for remagnetization.

Another fact that may have affected the scatter of A_R directions is the polarity. K.J. Neuvonen (personal communication, 1986) has observed that as in the Sipoo dykes, in the quartz porphyry dykes associated with the Wiborg rapakivi batholith, the scatter of reversed polarity directions is greater than that of the normal polarity. In Sipoo the age difference between diabases and quartz porphyries is very small, as shown by the often mixed and mingled magmas around the Subjotnian rapakivi granite and dyke areas in southern Finland (e.g. Rämö, 1991; Eklund, 1993). One possible cause for the scatter of A_R directions in the diabases is that some data points record a transitional field during a single polarity switch from reversed to normal. According to Hoffman (1993), during reversal when the geocentric dipole is weak, the magnetic poles do not

simply rotate from one polarity to the other, but instead, they appear to become significantly inclined from the geographic poles and remain there for long durations. In that case, however, it is expected that the transitional (i.e. intermediate between full stable polarity states) diabases would have markedly low remanence intensities, which was not observed.

10.2. Ages of poles

10.2.1. Pole A

In order to estimate the magnetization ages (poles) it is worthwhile to review the database (Torsvik et al., 1990; Pesonen et al., 1991; Elming et al., 1995) of other palaeomagnetic data of Subjotnian formations. A lot of palaeomagnetic data from Subjotnian formations have been obtained from Finland and Sweden (e.g. Neuvonen and Grundström, 1969; Neuvonen, 1970, 1986; Bylund, 1985; Pesonen et al., 1985, 1991; Pesonen, 1987; Bylund and Pesonen, 1987; Bylund and Elming, 1992). Most data are from rapakivi granites and associated diabase and quartz porphyry dykes. There are, however, not many palaeomagnetic poles of highest quality (class A–B after the modified Briden and Duff classification system (1981), see Pesonen et al., 1989) which carry well-dated isotopic ages (see Pesonen

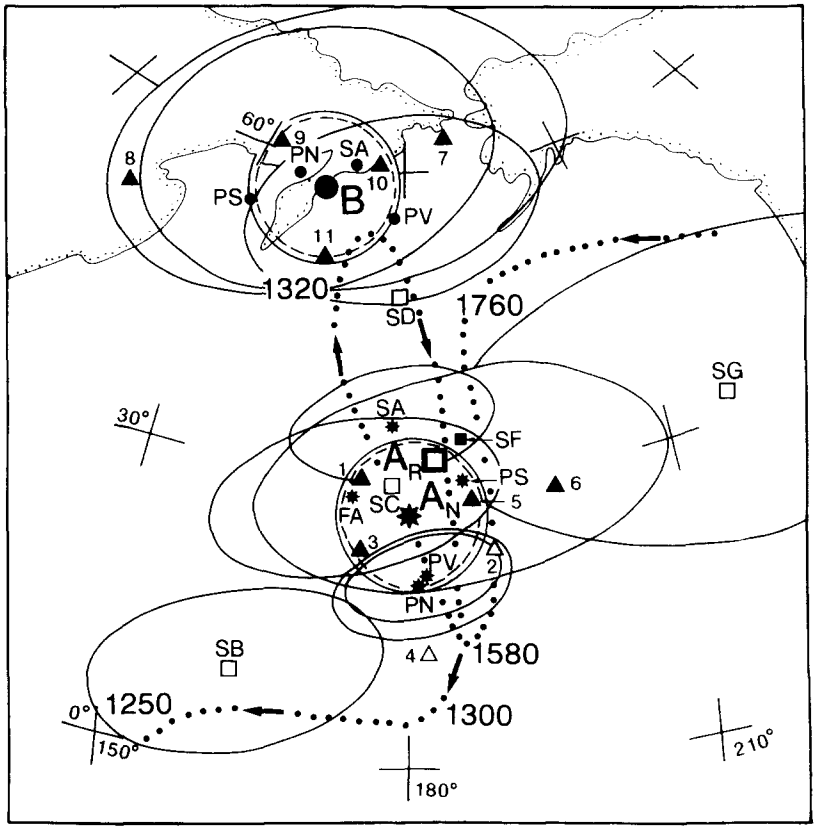


Fig. 17. Virtual geomagnetic poles (VGP) with confidence ovals for magnetizations A_N (stars) and B (circles) of quartz porphyry dykes and A_R (squares) of diabase dykes. The mean poles (A_N and B) for quartz porphyries are shown using larger symbols with their $\alpha 95$ confidence circles. The mean pole A_R of diabases is shown as a large open square without confidence oval. Other poles (triangles) are referred to in the text and in Fig. 1 and Table 5. Closed (open) symbols denote normal (reversed) polarity. The APWP segment (dotted curve) is based on an updated palaeomagnetic database for Fennoscandia (Elming et al., 1993b).

et al., 1991). However, there are some well-dated palaeopoles which can be directly correlated with the data obtained from the Sipoo dykes.

The purpose in correlating primary remanences and their pole positions, is to determine relative block movements between the different parts of the shield. The other goal is to find out if the secondary remanence components have regionally affected larger areas or if they are more local. In the following, Subjotnian poles with palaeomagnetic grade A or B are compared with poles obtained from Sipoo dykes (Fig. 17, Table 5). The relative ages of different magnetizations defined in the ca. 1.63–1.65 Ga old Sipoo dykes, cannot be uniquely established based only on the APW Path, because the Subjotnian loop closes itself at ca. 1.3 Ga.

The Sipoo A_N pole, regarded as primary in the quartz porphyries of age 1633 Ma (U–Pb zircon age, Törnroos, 1984) corresponds to the normal polarity pole (Pole 1, Fig. 17) of the quartz porphyry dykes (Neuvonen, 1986) associated with the Wiborg rapakivi granite. A reversed polarity, highly scattered remanence direction also occurs in these dykes (K.J. Neuvonen, personal communication, 1986). The quartz porphyry dykes have a U–Pb zircon age of 1617–1638 Ma and are of palaeomagnetic grade B. The isotopic age coincides with the age of the Sipoo quartz porphyries and thus these dykes, based on palaeomagnetic and isotopic data are coeval and have been magnetized at the same time about 1630 Ma ago. Neuvonen (1978) has isolated a similar but reversed polarity grade A pole from the contact zone of the Kuisaari dolerite (Pole 2, Fig. 17) which is baked by the Ahvenisto rapakivi pluton (age 1650 Ma; Vaasjoki, 1977) associated with the Wiborg batholith. This pole is considered to represent the age of the Ahvenisto rapakivi, 1650 Ma.

In western Fennoscandia, the normal polarity volcanic Upper Dala porphyries in southern Sweden (Fig. 1) carry also a magnetization in accordance with the A_N pole from Sipoo quartz porphyries (Pole 3, Fig. 17) (Mulder, 1971). These rocks have a Rb–Sr whole-rock age 1634 ± 38 Ma (Welin and Lundqvist, 1970), which is probably younger than the emplacement age. However, pole A_N is also close to the pole of the younger reversed polarity pole from the 1550–1601 Ma old (U–Pb zircon and Rb–Sr whole-rock ages, Suominen, 1987; Pesonen et al., 1985) Kumlinge–Brändö dykes (Pole 4, Fig. 17) and to the normal polarity pole of the Föglö–Sottunga dykes (Pole 5, Fig. 17) of age 1540 ± 12 Ma (U–Pb zircon age, Suominen, 1991) in the Åland archipelago of southwestern Finland (Pesonen and Neuvonen, 1981). The Åland dykes yielded a positive baked contact test and the magnetization is thus regarded as primary (grade A). Primary intrusion age was also suggested for the normal polarity pole of the 1510–1560 Ma old (Rb–Sr age; Patch-

Table 5
Palaeomagnetic poles used for comparison in Fig. 17.

Pole no.	Rock formation	Palaeomagnetic pole		Polarity	*dp/dm* (°)	Age (Ma)	Reference
		Lat. (°N)	Long. (°E)				
1	SE quartz porphyry dykes	30.0	175.0	N	6.6/13.3	1617–1638	Neuvonen (1986)
2	Kuisaari dolerite	22.0	190.0	R	7.0/14.0	1670–1630	Neuvonen (1978)
3	Upper Dala porphyries	22.7	175.6	N	21.6/42.9	1634 ± 38	Mulder (1971)
4	Kumlinge-Brändö dykes	12.2	182.0	R	5.0/ 9.0	1550–1601	Pesonen and Neuvonen (1981)
5	Föglö–Sottunga dykes	27.9	187.5	N	6.0/13.0	1523–1553	Pesonen and Neuvonen (1981)
6	Bräven–Hälleforsnäs dykes, EW1	27.7	196.8	N	4.1/ 8.2	1510–1560	Bylund (1985)
7	Dala Floda granites	64.0	189.0	N	7.0/10.0	–	Bylund and Elming (1992)
8	Dala, Gutsåvägen porphyries	52.0	133.0	N	–	–	Bylund and Elming (1992)
9	Dala, Fiskarbo dolerites	62.0	152.0	N	7.0/10.0	–	Bylund and Elming (1992)
10	Bräven-Hälleforsnäs dykes, EW2	61.7	176.1	N	9.3/13.7	1530	Bylund (1985)
11	Ragunda gabbro-anorthosite	51.6	166.6	N	5.7/ 8.9	1292 ± 30	Piper (1979), Pesonen et al. (1991)

ett, 1978) east–west-trending Breven and Hälleforsnäs olivine dolerite dykes (Pole 6, Fig. 17) in southern Sweden (Bylund, 1985).

The virtual geomagnetic poles (VGP) of the reversed polarity Sipoo diabases are highly scattered, but the mean pole is close to the normal polarity pole obtained from the quartz porphyries. In Sipoo, it is not possible to determine the magnetization age of the diabases based on the time of the polarity change of the reversed polarity diabases to the normal polarity quartz porphyries, because the time duration of the reversal process is geologically so rapid, less than 104 years (Tarling, 1983). The frequency of polarity reversals is not well known, especially in the Precambrian, but the polarity can remain constant for hundreds of thousands to million years (Tarling, 1983). However, since the normal polarity secondary overprint (A_N), whose direction corresponds to that in the quartz porphyries, has been found in the most altered diabase dykes, it shows that in the diabases the primary magnetization (A_R) is clearly older than that (A_N) in the quartz porphyries.

These data show that the same magnetization, regarded as primary Subjotnian remanence in all the formations studied, has been found in rocks of similar age in a wide zone from Wiborg rapakivi batholith in the east through Sipoo dykes in the middle and to the Dala porphyries in the west (Fig. 1). The result supports the conclusion that the southern Fennoscandian shield has been stable without large-scale intraplate tectonic movements since Subjotnian times.

10.2.2. Pole B

The high-coercivity component *B* in Sipoo quartz porphyries has been isolated as a secondary magnetization in many other Subjotnian formations in Sweden and Finland. Bylund and Elming (1992) report a corresponding component with steep positive inclination of high coercivity and high blocking temperature in Dala granites (Pole 7, Fig. 17), porphyries (Pole 8, Fig. 17) (Rb–Sr whole-rock age about 1650 Ma) and dolerites (Pole 9, Fig. 17) in central Sweden. In addition, Bylund (1985) has isolated a similar magnetization in the 1510–1560 Ma old east–west-trending Breven and Hälleforsnäs dykes (Pole 10, Fig. 17). The origin of this secondary remanence was not resolved in either of those formations but, based on their high coercivities and high unblocking temperatures, they were not regarded as purely viscous although the direction is very close to the PEF. Corresponding palaeopoles have been reported from Sweden also earlier, but most of their ages have not been confirmed by isotopic age data. For example, Piper (1979) reports a similar pole from the Ragunda gabbro-anorthosites (Pole 11, Fig. 17) which has been dated as 1320 ± 30 Ma (Kornfält, 1976) and 1291 ± 30 Ma old (recalculated age in Piper, 1979). The pole is palaeomagnetic grade A. In Finland, a similar high-coercivity intermediate direction remanent magnetization has been isolated in Subjotnian Åland dykes (Pesonen et al., 1985) and in 1880 Ma old Keuruu dykes (L.J. Pesonen, unpublished data, 1993).

The pole position of the high-coercivity component *B* (Fig. 17), although probably not adequately isolated, implies an age of about 1320 Ma, defined purely on the APWP of Fennoscandia. The time between 1500 and 1250 Ma is known to be geologically relatively quiet without any significant tectonic and magmatic events in the Fennoscandian shield (Gorbatschev et al., 1987). However, drift and rotational velocities for the whole Fennoscandian shield (Pesonen et al., 1989; Elming et al., 1993) have a peak at about 1.45–1.25 Ga ago. Different hypotheses have been suggested to explain this phenomenon. Pesonen et al. (1989) suggested a local hot spot, over which the Fennoscandian plate has drifted and to which high velocities may be related. Elming et al. (1993) suggest that increased APW drift rates at about 1.3 Ga ago could be explained by splitting up of a larger continent, including the Ukrainian plate existing in the Early–Middle Proterozoic. Park (1993) also has suggested, based mainly on palaeomagnetic data of Pesonen et al. (1989) and Piper (1982), that the time about 1.3 Ga ago can mean the beginning of the breakup of the Proterozoic supercontinent (Fennoscandia and Laurentia). The dyke swarms emplaced at about 1.26 Ga ago, for example the Satakunta Postjotnian dykes in the Fennoscandia (Fig. 1)

and the McKenzie dyke swarm in the Canadian shield, are a world-wide expression of extensional rift related tectonics at that time. Rämö (1990, 1991) suggests that the Subjotnian dykes were formed as a result of crustal melting caused by large-scale mantle upwelling owing to formation of the supercontinent Laurentia and that later, about 1.26 Ga ago, the Jotnian formations in western Finland were probably formed in a passive rift which may be related to the onset of the Sveconorwegian–Grenvillian orogeny in southern Fennoscandia.

Thus a hypothesis of a large-scale tectonic event, either a break up or a collision of continents about 1300–1200 Ma ago has to be taken into account in attempts to explain remanence *B*. A concordia diagram of U–Pb analysis of zircon for a quartz porphyry dyke in Pellinge, close to the Sipoo dykes gives the upper intercept age of 1633 ± 10 Ma representing the emplacement age. The lower intercept gives an age of 183 ± 118 Ma (Vaasjoki et al., 1992) which can be regarded as geologically meaningless. In addition, Suominen (1991) has shown that in southwestern Finland, in the Åland rapakivi suite and in the Svecofennian rocks, the lower intercept ages are not higher than about 660 Ma. Also in southeastern Finland, in the Wiborg rapakivi batholith, the highest lower intercept age is 414 Ma (Vaasjoki et al., 1989). On the other hand, in Grenville Province, Canada, Connelly and Heaman (1993) report upper intercept ages (U–Pb analysis on zircon and monazite) of approximately 1630 Ma for granitoids and migmatites formed during the Labradorian orogeny and lower intercept ages of about 1000 Ma. The lower intercept ages are interpreted to result from the Grenvillian orogeny about 1200–1000 Ma ago, and the slight Pb-loss in zircon and monazite, shown as lower ages, is due to considerable heating during the orogeny. In the Sipoo dykes, a reason for the absence of isotopic resetting may be that the thermal effects were not efficient enough to reset the U–Pb system. This supports the view that hydrothermal fluids were responsible for the acquisition of component *B*. The only isotopic age evidence on lower ages in the area comes from hydrothermally altered rocks in a fracture zone north of Helsinki, near Sipoo where Jäger and Pipping have got a Rb–Sr isochron age of 1318 ± 54 Ma (F. Pipping, personal communication, 1993). As has been shown in some recent studies (e.g. Oliver, 1986) hydrothermal fluids released from orogenic belts may remagnetize rocks far away from the orogenic belt itself. In southern Fennoscandia, the onset of the Sveconorwegian–Grenville orogeny or the earlier separation of Laurentia from Fennoscandia may have contributed to the acquisition of magnetization *B* at about 1320 Ma ago. The occurrence of this hard magnetization is areally relatively widely spread both in Finland and Sweden and thus a common source is obvious. However, a more thorough understanding of the processes and mechanisms which have produced the secondary overprint still needs more accurate palaeomagnetic work over large areas, combined with sophisticated mineral identification and physical property determination methods.

11. Conclusions

The following conclusions can be drawn from the palaeomagnetic study of the Sipoo dykes.

(1) The Sipoo quartz porphyry dykes are characterized by SD/PSD magnetite grains. The intermediate- to high-coercivity normal polarity A_N component is considered to be a TRM acquired during the primary cooling of the dykes about 1630 Ma ago. The pole position of A_N corresponds to poles obtained from different formations of similar age in southern Sweden and Finland and indicates that there have not been any vast tectonic movements between different blocks in the shield since that time.

(2) The diabases, characterized by large MD magnetite grains, carry an intermediate- to high-coercivity, reversed polarity magnetization A_R nearly antiparallel to magnetization A_N in the quartz porphyries. Based on a partial contact test A_R is a primary TRM, but owing to scattered data its magnetization age cannot be determined.

(3) As a consequence of hydrothermal fluids released from the rapakivi granites and quartz

porphyries, the diabases have been partially remagnetized and carry a secondary chemical remanence whose direction corresponds to the primary remanence A_N in the quartz porphyries. The petrographically most altered dyke which also shows lowered susceptibility and remanence intensity values, is almost completely remagnetized. In the best preserved diabase dyke the partial remagnetization is also detectable as a high-coercivity component carried by newly formed magnetite, in addition to the primary component. This result shows that there is a clear age difference between the reversed (older) and normal (younger) polarity magnetizations.

(4) In the quartz porphyries there is also a high-coercivity normal polarity component B which can be a combination of the A_N and PEF components with no geological meaning. Because a similar high-coercivity and high blocking temperature component has also been isolated in different rock types in southern Sweden and Finland, it is suggested that B represents a chemical overprint related to the reactivation of the whole southern Fennoscandian shield about 1320 Ma ago. This age for B magnetization is based on its pole position on the Fennoscandian APW path although the age does not correlate with any known tectonothermal or structural events in the area. It is proposed that it may have been acquired in the break up of Laurentia or at the onset of the Sveconorwegian–Grenvillian orogeny.

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